1 Data and supervision

3328 frames, 550×550 +flip



by image id, no overlap

1664 | 208 | 208

ReconIoU



- ✓ 0.889
- ✓ 0.890
- ✓ 0.891

λ_{mask} 1.0 BCE+Dice

λ_{par} 0.5 L1

λ_{lm} 1.0 generation

Supervise

2 Parallel trainable branches

Vision encoder
2 layers d=1152

LLM decoder
24 layers d=1536

SAM2 mask decoder

first [SEG] hidden state

base LLM frozen + LoRA A:B

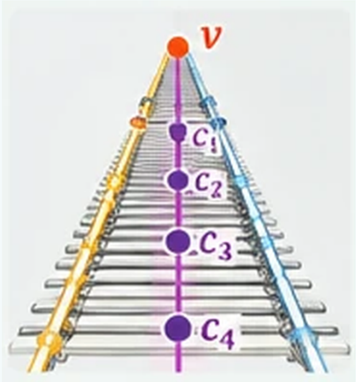
r=128, α=256 on attn.+MLP

f_ψ 3× Linear
1536, 2x ReLU, zero-init, hand-set bias

2	8	4	4	2
v	$c_{1:K}$	$w^L_{1:K_w}$	$w^R_{1:K_w}$	t

$\phi \in \mathbb{R}^{20}$

Rail track parameterization



Differentiable rasterizer

1. Densify $K = 4 \rightarrow 32$
2. Side = sign(cross)
3. e = $\tau(\ell - d)$, $\tau = 2$
4. $M = \sigma(e)$



Backpropagate to trainable branches

- 2: v vanishing point (2-D)
- 8: $c_{1:K}$ centerline control points ($K = 4 \times 2$)
- 4: $w^L_{1:K_w}$ left width samples
- 4: $w^R_{1:K_w}$ right width samples
- 2: t topology logits (for switches)